

Wai Shing Tang, Ph.D.

Email: wstangpp@gmail.com | Website: wtang8.github.io

AI for Science • Generative Models • Bayesian Learning • Scientific Machine Learning

PROFESSIONAL EXPERIENCE

Independent AI Scientist & Research Engineer Consultant

Feb 2026 – Present

Self-Employed

Remote

- Work with clients on AI research, machine learning engineering, and scientific data modelling projects across domains.
- Developed a prototype multi-modal transformer-based foundation model for cross-domain prediction and generation tasks.
- Designed data pipelines, tokenization strategies, and deep learning components in PyTorch and JAX.
- Developed uncertainty-aware dynamical models and Bayesian optimization algorithms for real-world telemetry and physical data analysis, integrating Bayesian inference, control theory, and machine learning.
- Applied statistical modelling and machine learning methods to extract insights from complex, sparse, and high-dimensional temporal datasets to support data-driven decision-making.

Senior Researcher, AI4Science

May 2025 – Oct 2025

Microsoft Research

Cambridge, UK

- Designed Bayesian inference frameworks and probabilistic models to incorporate large-scale, noisy cryo-EM data into diffusion generative models for protein structure prediction.
- Collaborated with machine learning researchers and software engineers to implement scalable scientific computing workflows.

Flatiron Research Fellow

Jun 2022 – Apr 2025

Flatiron Institute (CCM & SMBp)

New York City, NY, USA

- Led the development of Bayesian inference frameworks for extracting molecular structural information from highly noisy cryo-EM imaging data.
- Optimized likelihood-based inference algorithms using PyTorch and GPU-accelerated computing, achieving more than 20× acceleration over CPU baselines, enabling new Bayesian applications in biomolecular structure determination.
- Integrated the algorithms into end-to-end software pipelines for processing and analyzing hundreds of TB of cryo-EM datasets across multiple GPUs on HPC.
- Extended Bayesian inference approaches beyond cryo-EM to experimental chromatin datasets, developing methods to infer ensemble structural distributions from heterogeneous biological measurements.
- Mentored junior researchers on Bayesian modelling, scientific machine learning, molecular dynamics simulations, and high-performance computational workflows.

Ph.D. Researcher

Sept 2017 – May 2022

Brown University

Providence, RI, USA

- Developed computational methods combining molecular simulation, statistical inference, and machine learning to study complex biomolecular systems.
- Performed large-scale molecular dynamics simulations (using GROMACS and OpenMM) to investigate molecular interactions, conformational landscapes, and phase behavior of intrinsically disordered proteins.
- Applied advanced enhanced sampling methods, including REMD (Replica Exchange Molecular Dynamics) and PT-WTE (Parallel Tempering Well-Tempered Ensemble) to characterize rare conformational states and molecular mechanisms.
- Developed Gaussian process-based feature selection and topological data analysis approaches to extract meaningful structural representations and identify structural features distinguishing protein variants and predict mutational effects from short (<200 ns) MD simulation trajectories.
- Built scalable HPC workflows for molecular simulation, trajectory analysis, and large-scale scientific data processing.

EDUCATION

Ph.D. in Physics

Brown University

Sept 2017 – May 2022

Providence, RI, USA

- **Advisor:** Lorin Crawford
- **Thesis:** Topological Data Analysis framework for biological structures and short-term evolving molecular models

B.Sc. in Physics (Enrichment Stream in Physics)

The Chinese University of Hong Kong

Sept 2013 – Jun 2017

Hong Kong SAR

SOFTWARE DEVELOPED

1. **SINATRA Pro:** Protein Structure and Conformation Analysis using Topological Summary Statistics (Python and C/C++).
github.com/lcrawlab/SINATRA-Pro
2. **Ensemble Reweighting using Cryo-EM Particles:** Statistical reweighting framework for structural ensembles.
github.com/flatironinstitute/Ensemble-reweighting-using-Cryo-EM-particles
3. **CryoLike:** Python package for Cryo-EM image-to-structure likelihood calculations.
github.com/flatironinstitute/CryoLike

SELECTED PUBLICATIONS

1. **W. Tang**, J. Soules, A. Rangan, P. Cossio (2025). CryoLike: A Python package for cryo-electron microscopy image-to-structure likelihood calculations. *Acta Cryst. D*.
2. **W. Tang**, D.S. Sánchez, J.G. Barreto, B. Carpenter, S.M. Hanson, A.H. Barnett, E.H. Thiede, P. Cossio (2023). Ensemble reweighting using Cryo-EM particles. *Journal of Physical Chemistry B*, 127(24), 5410–5421.
3. **W. Tang**, E.D. Zhong, S.M. Hanson, E.H. Thiede, P. Cossio (2023). Conformational heterogeneity and probability distributions from single-particle cryo-electron microscopy. *Current Opinion in Structural Biology*, 81, 102626.
4. **W. Tang**, G.M. da Silva, H. Kirveslahti, E. Skeens, B. Feng, T. Sudijono, K.K. Yang, S. Mukherjee, B. Rubenstein, and L. Crawford (2022). A topological data analytic approach for discovering biophysical signatures in protein dynamics. *PLOS Computational Biology*, 18(5): e1010045.
5. A.C. Murthy, **W. Tang**, N. Jovic, A. Janke, D.H. Seo, T. Perdikari, J. Mittal, and N.L. Fawzi (2021). Molecular interactions contributing to FUS SYGQ LC/RGG phase separation and co-partitioning with RNA polymerase II heptads. *Nature Structural & Molecular Biology*, 28, 923–935.
6. **W. Tang**, N.L. Fawzi, and J. Mittal (2020). Refining All-Atom Protein Force Fields for Polar-Rich, Prion-like, Low-Complexity Intrinsically Disordered Proteins. *Journal of Physical Chemistry B*, 124(43): 9505-9512.
7. B.S. Schuster, G.L. Dignon, **W. Tang**, F.M. Kelley, A.K. Ranganath, C.N. Jahnke, A.G. Simpkins, R.M. Regy, D.A. Hammer, M.C. Good, and J. Mittal (2020). Identifying sequence perturbations to an intrinsically disordered protein that determine its phase-separation behavior. *PNAS*, 117(21): 11421-11431.